IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies
  - The project was completed through these steps:
    - Data Collection
    - Data Wrangling
    - Exploratory Data Analysis
    - Interactive Visual Analytics
    - Predictive Analysis
- Summary of all results
  - The following results were produced:
    - Exploratory Data Analysis results
    - Geospatial Visualization with Folium
    - Interactive Dashboard with Plotly
    - Predictive Analysis results

Introduction

- SpaceX sends manned missions to Space. One reason SpaceX can do this is the rocket launches are relatively inexpensive. SpaceX advertises Falcon 9 rocket launches on its website with a cost of 62 million dollars. SpaceX can do this for less than other companies because SpaceX can reuse the first stage.
- Space Y would like to compete with SpaceX and needs to know the price of each launch to determine if Space Y wants to bid against SpaceX for a rocket launch.
- If we can determine if the first stage will land, we can determine the cost of a launch. We will train a machine learning model and use public information to predict if SpaceX will reuse the first stage.

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
    - Data were gathered from the SpaceX REST API
- Perform data wrangling
    - Null values were removed
    - A new class was created to show if the outcome of the launch was successful
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
    - Data were split into training and testing sets
    - Multiple models were trained and evaluated using the training and testing data

Data Collection – SpaceX API

- A GET request was used to request and parse the SpaceX launch data
- The dataframe was filtered to only include Falcon 9 launches
- The .mean() and .replace() functions were used to replace missing values for the Payload Mass with the mean
- https://github.com/daniellerrussell/IBMdatascience/blob/main/jupyter-labs-spacex-data-collection-api(2).ipynb

Request and parse the data

- Use a GET request
- Decode the response content as a JSON
- Turn it into a Pandas dataframe

Filter the dataframe

- Keep only the Falcon 9 launches
- Save the filtered data to a new dataframe
- Reset the Flight Number column

Data Wrangling

- Calculate the mean value of Payload Mass
- Replace the null values with the mean

Data Collection - Scraping

- Falcon 9 launch records were web scraped with Beautiful Soup
  - Requested the Falcon 9 Launch Wiki page
  - Extracted column/variable names
  - Created a data frame by parsing the launch HTML tables
- https://github.com/daniellerrussell/IBMdatascience/blob/main/jupyter-labs-webscraping.ipynb

Request the HTML page and get a response object

- Perform an HTTP GET method
- Create a BeautifulSoup object from the HTML response

Extract column names from the HTML table header

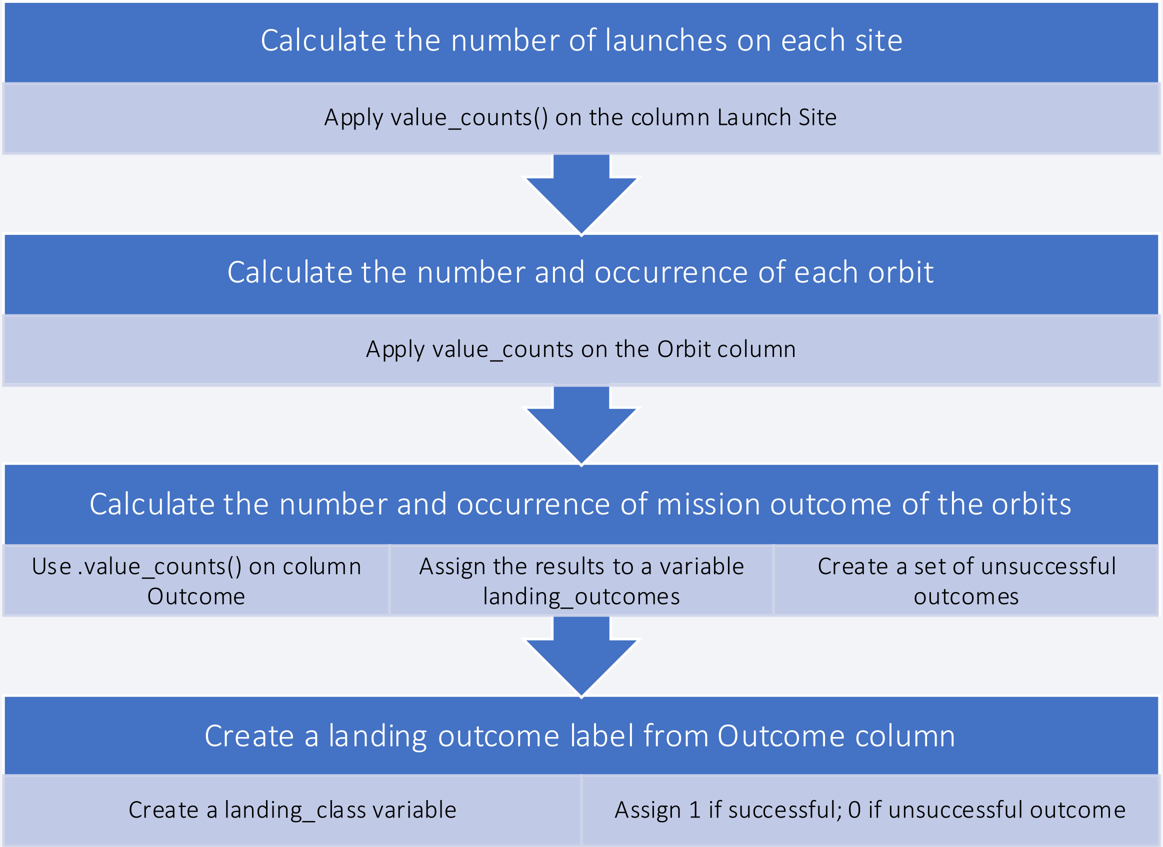
- Iterate through <th> elements
- Apply extract_column_from_header()

Create a dataframe by parsing the launch HTML tables

- Create a launch dictionary
- Fill the launch dictionary with extracted rows
- Create a dataframe from the launch dictionary

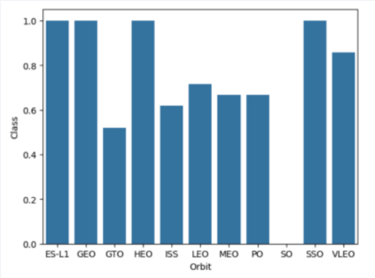
Data Wrangling

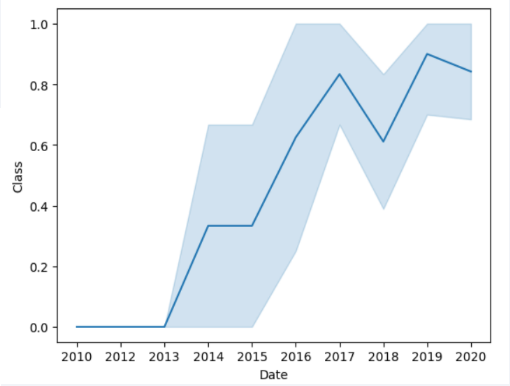
- First, the number of launches on each site was calculated using the method value_counts() on the column LaunchSite
- The method value_counts was also used to calculate the number and occurrence of each orbit
- The number and occurrence of mission outcome of the orbits was also calculated
- A landing outcome label was created
- https://github.com/daniellerrussell/IBMdatascience/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb

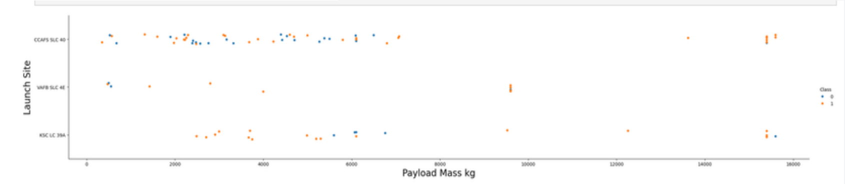


EDA with Data Visualization

- The function catplot was used to create scatter plots to visualize the relationship between:
    - Flight Number and Launch Site
    - Payload Mass and Launch Site
    - Flight Number and Orbit Type
    - Payload Mass and Orbit Type







- The groupby method was used on the Orbit column with the mean of the Class column to create a bar chart to visualize the success rate of each orbit
- A function to extract the year from the date was used to create a line chart to visualize the success rate over time
- https://github.com/daniellerrussell/IBMdatascience/blob/main/edadataviz.ipynb

EDA with SQL

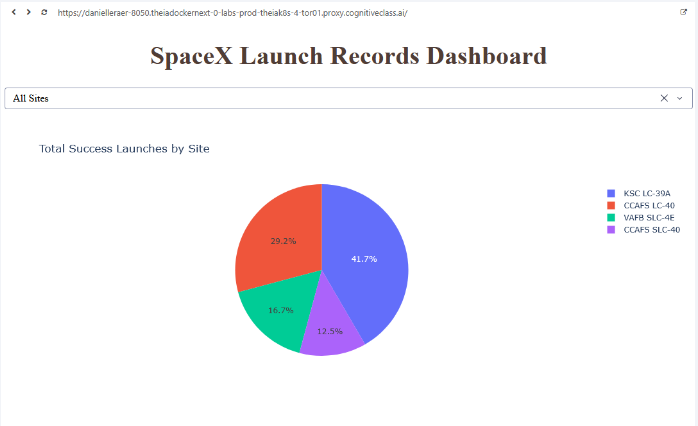
- SQL queries were performed to:
  - Display the names of the unique launch sites
  - Find unique launch site names and records for specific launch sites
  - Display the total payload mass and average payload mass
  - List the date when the first successful landing outcome in ground pad was achieved
  - List the names of boosters with specific success criteria
  - List the total number of successful and failure mission outcomes
  - List the booster_versions that have carried the maximum payload mass
  - List the records with specific criteria in 2015
  - Rank the count of landing outcomes in descending order
- https://github.com/daniellerrussell/IBMdatascience/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb

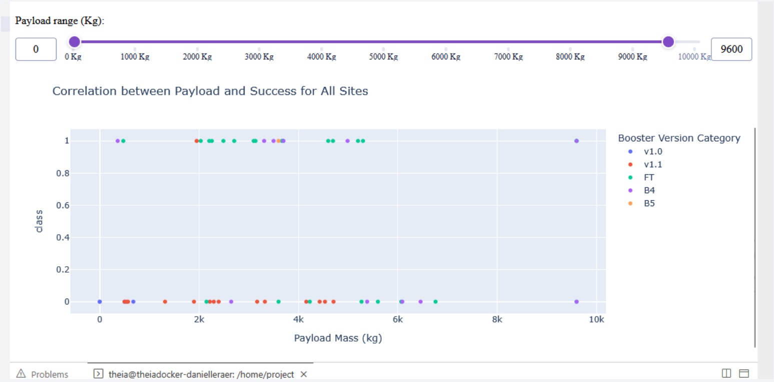
Build an Interactive Map with Folium

- Markers, circles, and lines were added to a Folium map to visualize geographical patterns about launch sites
- First, a circle and marker were added to the map for each launch site
- A marker cluster object was created to display the launch results for each site with a green marker if the launch was a success and a red marker for an unsuccessful outcome
  - The color-labeled markers in the marker clusters allow us to be able to easily identify which launch sites have relatively high success rates
- A MousePosition was added to the map to get coordinates for a mouse over a point on the map
  - This allows us to easily find the coordinates of any points of interest while we explore and analyze the proximities of launch sites
- PolyLines were drawn on the map to mark the distance between a launch site to a selected coastline point and highway point
  - Plotting distance lines to proximities allows us to visualize the distance between the launch sites and coastlines, highways, railways, cities, etc.
  - This helps us to identify what geographic features are relevant to launch site selection
- https://github.com/daniellerrussell/IBMdatascience/blob/main/lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

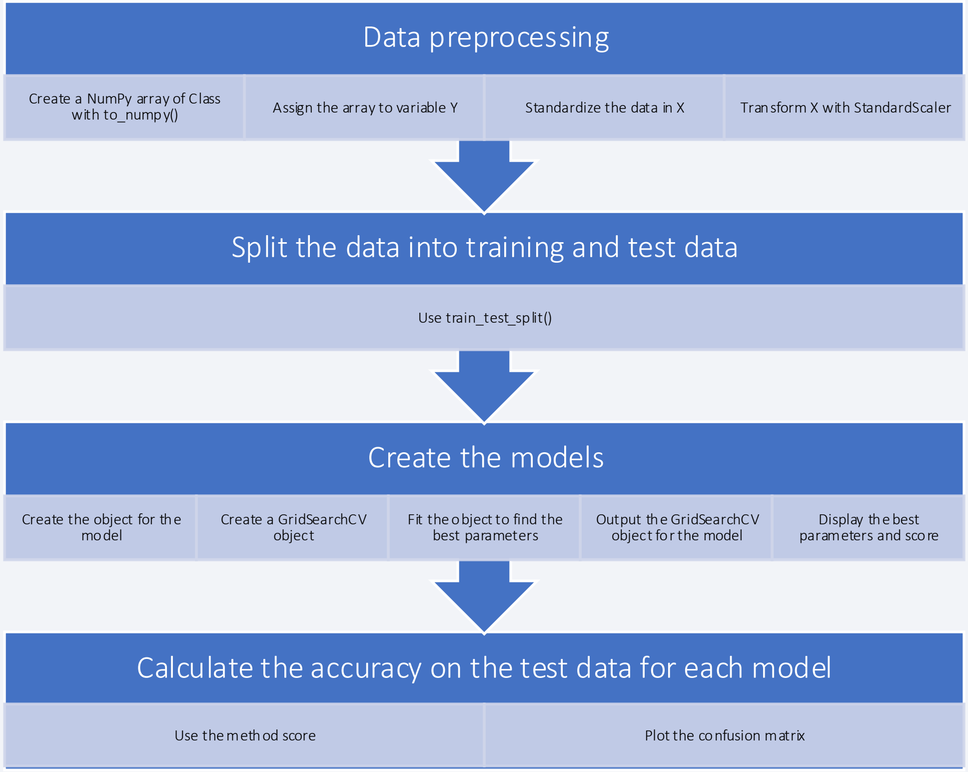
- A pie chart was added to the dashboard to show the success rate of launch sites
- A scatter plot was added to the dashboard to show the relationship between payload and launch success
- A dropdown was added to filter the pie chart and scatter plot based on the selected launch site
- A range slider was also added to filter the scatter plot by the payload range selected
- These plots and interactions allow you to visualize success rates of launch sites by various criteria and determine which sites have the most launch success and which payloads are correlated with success at each site
- https://github.com/daniellerrussell/IBMdatascience/blob/main/spacex-dash-app(9).py





Predictive Analysis (Classification)

- A NumPy array was created from the column Class with to_numpy() and assigned to the variable Y
- The data in variable X was transformed using preprocessing.StandardScaler()
- The data were then split into train and test sets using the function train_test_split
- Four models were then developed and tested:
    - Logistic Regression
    - Support Vector Machine (SVM)
    - Decision Tree Classifier
    - K-Nearest Neighbors (KNN)
- A confusion matrix was generated for each model and each model was scored for accuracy to determine which model performed the best
- https://github.com/daniellerrussell/IBMdatascience/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb



Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

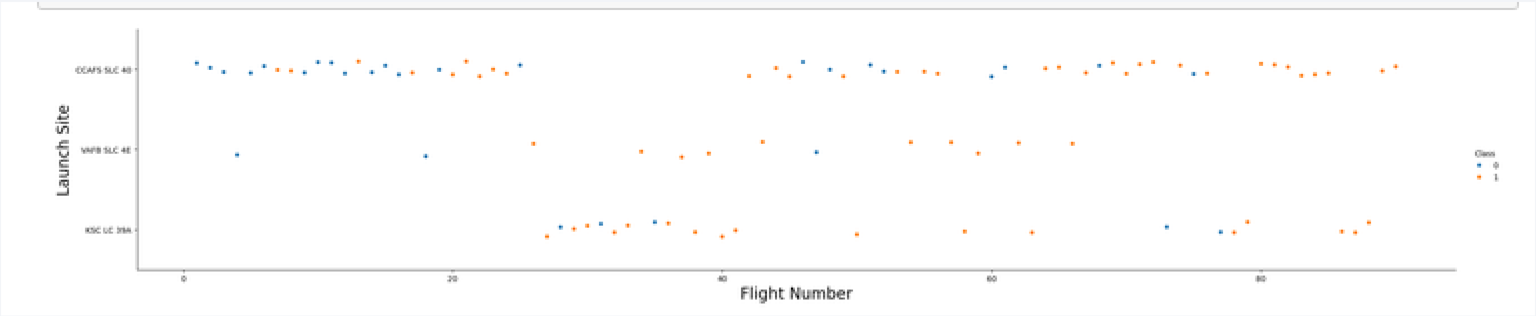
Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

The scatter plot of Flight Number vs. Launch Site shows:

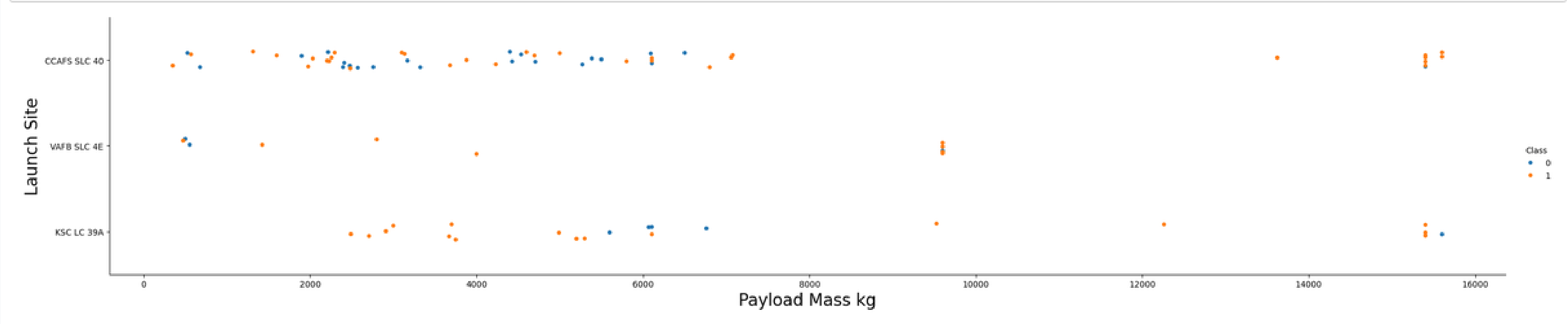
- As the flight number increases, the launch success rate increases
- Most early flights (flight numbers 25 and under) were launched from CCAFS SLC 40 and were mostly unsuccessful
- Flights from KSC LC 39A appear to be overall more successful than flights from CCAFS SLC 40; this may be partially due to the site not having early flights (numbers 25 and under)
- VAFB SLC 4e has the fewest flights, but still shows more unsuccessful launches at lower flight numbers



Payload vs. Launch Site

The scatter plot of Payload vs. Launch Site shows:

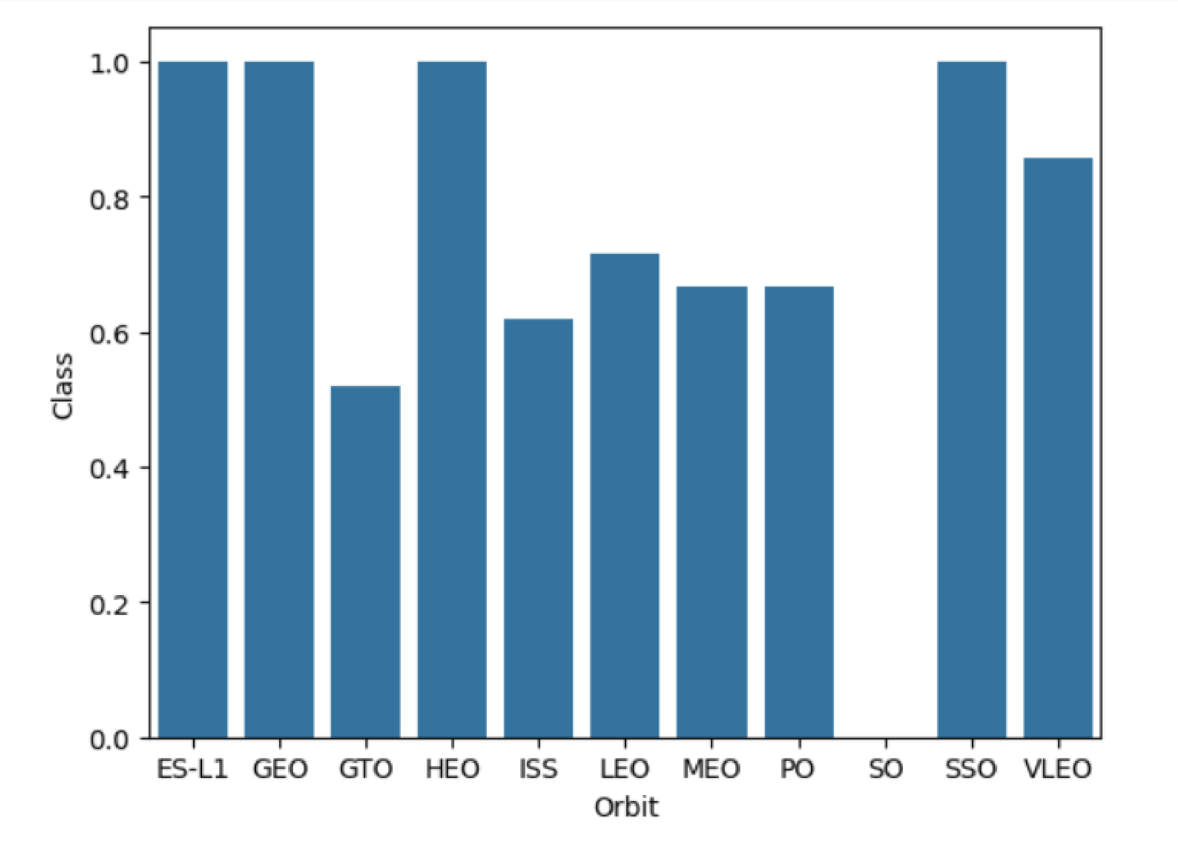
- The heaviest payloads launched from CCAFS SLC 40 and KSC LC 39A; no payloads over 10,000 kg launched from VAFB SLC 4E
- Most flights from CCAFS SLC 40 were payloads under 8,000 kg with mixed success
- There is less data for heavier launches (over 8,000 kg), but the success rate for heavy launches appears high



Success Rate vs. Orbit Type

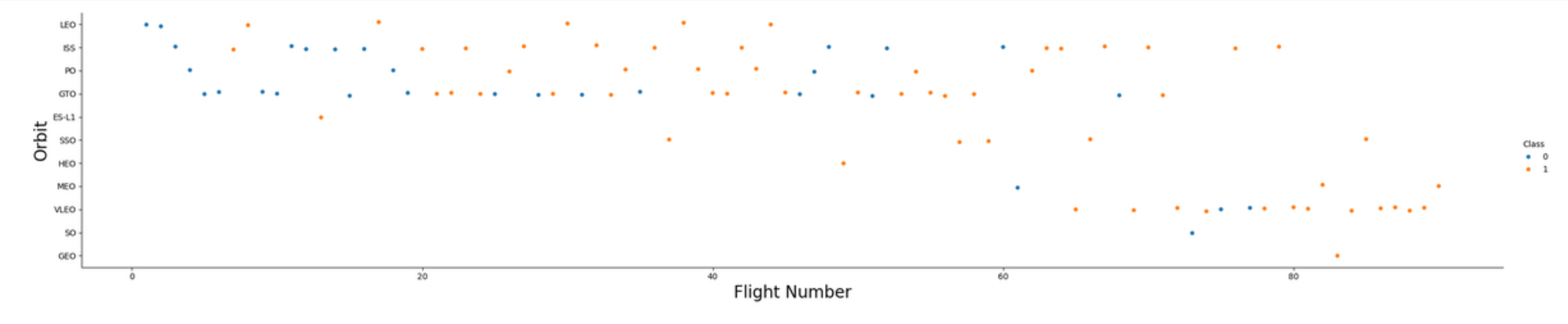
- The Success Rate vs. Orbit Type bar chart shows four orbit types have the highest success rate of 100%:
    - ES-L1
    - GEO
    - HEO
    - SSO

- The lowest success rate is SO with 0% success



Flight Number vs. Orbit Type

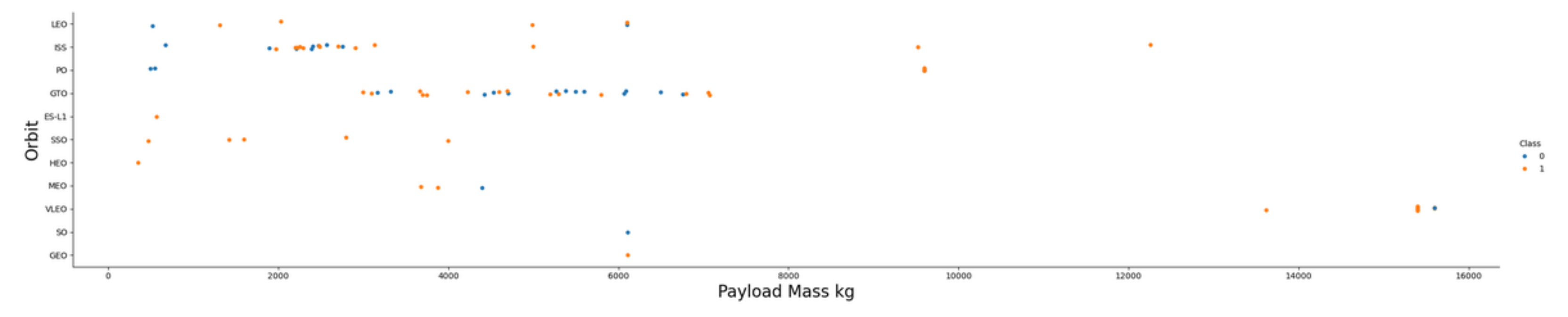
- The scatter point of Flight number vs. Orbit type shows:
- HEO, GEO, and ES-L1each only have one flight and it was successful, which explains why they have 100% success rates
- SO also has only one flight and it was unsuccessful, which explains why the success rate is 0%
- In the LEO orbit, success seems to be related to the number of flights
- Conversely, in the GTO orbit, there appears to be no relationship between flight number and success



Payload vs. Orbit Type

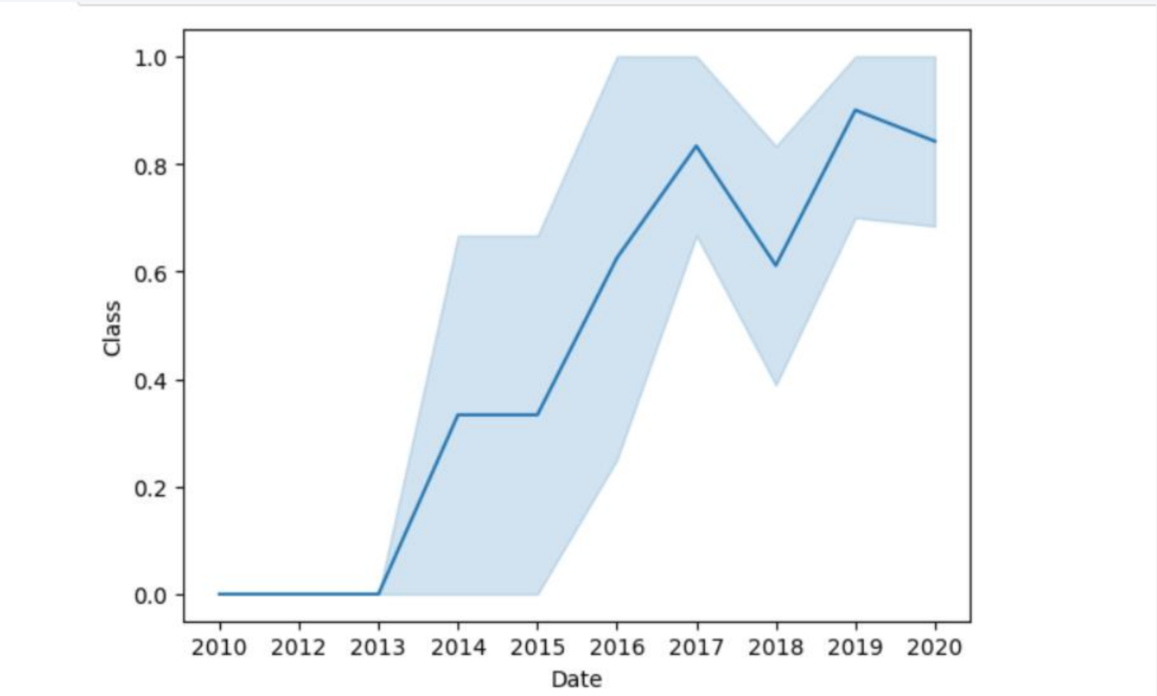
The scatter point of payload vs. orbit type shows:

- PO, ISS, and LEO have more success with heavy payloads
- For GTO, it's difficult to distinguish between successful and unsuccessful landings as both outcomes are present



Launch Success Yearly Trend

- The line chart of yearly average success rate shows:
- All landings were unsuccessful from 2010 to 2013
- After 2013, the success rate increases until 2020, with a dip in 2018



All Launch Site Names

- Adding DISTINCT to the select statement returns only distinct values from the Launch_Site column; there are four distinct launch sites

```
%sql select distinct "Launch_Site" from SPACEXTABLE
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- Using the LIKE operator with 'CCA%' keyword with wild card finds all records where the value in Launch_Site begin with `CCA`
- LIMIT 5 limits the response to 5 records

```
%sql select * from SPACEXTABLE where "Launch_Site" like 'CCA%' LIMIT 5
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- The SUM function calculates the total of the values in the Payload_Mass__KG column
- The LIKE operator with 'NASA%' keyword and wildcard on the Customer column limits the results to the values for customer NASA

```
%sql select sum("PAYLOAD_MASS__KG_") from SPACEXTABLE where Customer like 'NASA%'
```

sum("PAYLOAD_MASS__KG_")
99980

Average Payload Mass by F9 v1.1

- The AVG function calculates the average payload mass from the Payload_Mass__KG column
- Adding the LIKE operator in the WHERE clause filters the results to booster version F9 v1.1

```
%sql select avg("PAYLOAD_MASS__KG_") from SPACEXTABLE where "Booster_Version" like 'F9 v1.1'
```

avg("PAYLOAD_MASS__KG_")
2928.4

First Successful Ground Landing Date

- Using the MIN() function on the Date column finds the date of the first successful landing outcome
- Using the LIKE operator in the WHERE clause filters the results for only successful ground pad values in the Landing_Outcome column

```
%sql select min(Date) from SPACEXTABLE where "Landing_Outcome" like 'Success (g%'
```

min(Date)
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

- Using the LIKE operator with 'Success (drone%') keyword and wildcard on the Lnading_Outcome column in the WHERE clause filters the results to boosters which have successfully landed on drone ship
- Using the comparison operators ">" and "<" filter for a payload mass greater than 4000 but less than 6000

```
%sql select "Booster_Version" from SPACEXTABLE where ("Landing_Outcome" like 'Success (drone%') and
("PAYLOAD_MASS__KG_" > 4000) and ("PAYLOAD_MASS__KG_" < 6000)
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- The COUNT function calculates the total number of successful and failure mission outcomes

```
%%sql select count(case when "Mission_Outcome" like 'Success%' then 1 else 0 end) as 'Success',
count(case when "Mission_Outcome" like 'Failure%' then 1 else 0 end) as 'Failure'
from SPACEXTABLE
```

Success	Failure
101	101

Boosters Carried Maximum Payload

- A subquery with a MAX aggregate function is used to list the names of the booster which have carried the maximum payload mass

Booster_Version	PAYLOAD_MASS__KG_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

```
%%sql select distinct "Booster_Version", "PAYLOAD_MASS__KG_" from SPACEXTABLE
left join (select max("PAYLOAD_MASS__KG_") as maxkg from SPACEXTABLE) mkg on mkg.maxkg = SPACEXTABLE."PAYLOAD_MASS__KG_"
where SPACEXTABLE."PAYLOAD_MASS__KG_" = mkg.maxkg
```

2015 Launch Records

- The LIKE operator with 'Failure (d%' keyword in the WHERE clause is used to filter the results to failed landing_outcomes in drone ship
- The SUBSTR function is used to get the 'Month' from Date
- The WHERE clause also contains the comparison operator "=" to limit the year

```
%%sql select substr(Date, 6, 2) as 'Month', "Landing_Outcome", "Booster_Version", "Launch_Site"
from SPACEXTABLE where "Landing_Outcome" like 'Failure (d%' and substr(Date, 0, 5) = '2015'
```

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- The COUNT function, GROUP BY, and ORDER BY are used to rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

```
%%sql select "Landing_Outcome", count("Landing_Outcome") from SPACEXTABLE
where Date between '2010-06-04' and '2017-03-20'
group by "Landing_Outcome"
order by count("Landing_Outcome") desc
```

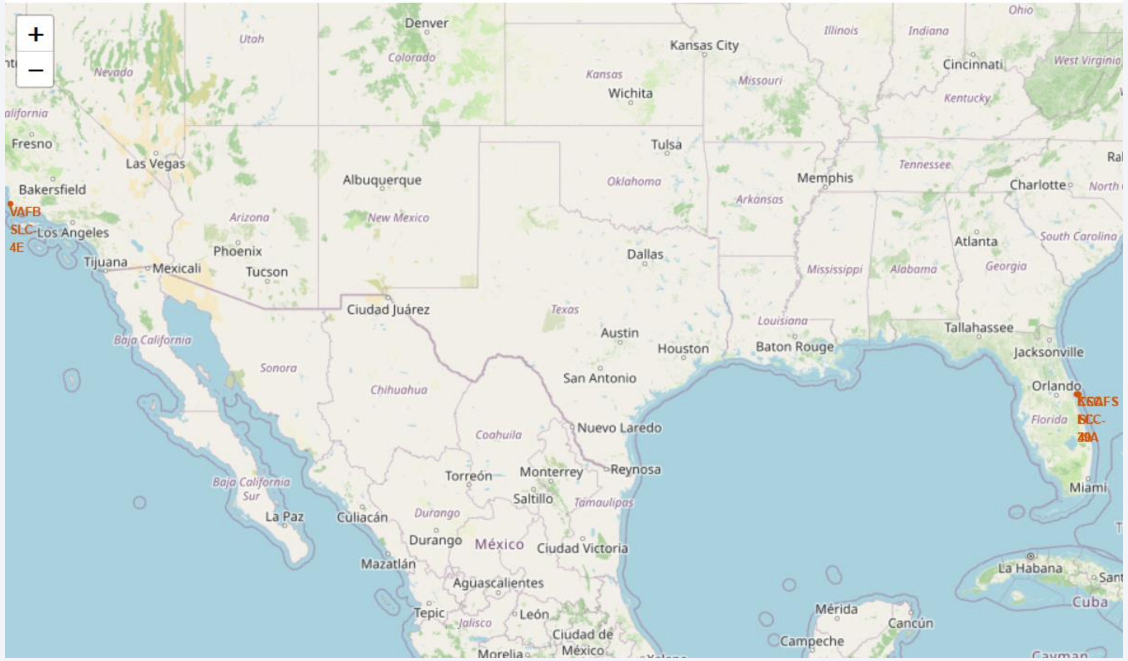
Landing_Outcome	count("Landing_Outcome")
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Section 3

Launch Sites Proximities Analysis

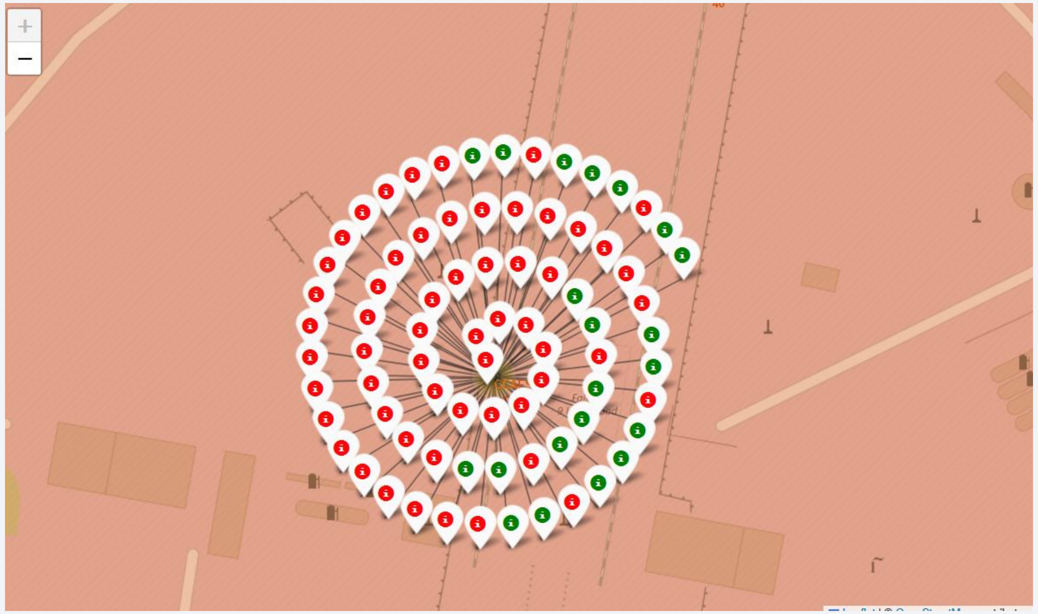
All Launch Sites Markers Folium Map

- The generated folium map shows all launch sites' location markers on a global map
- The markers are labeled with the site codes
- The map shows that launch sites are located on coastlines
- The map also shows that launch sites are located in the southern half of the continent



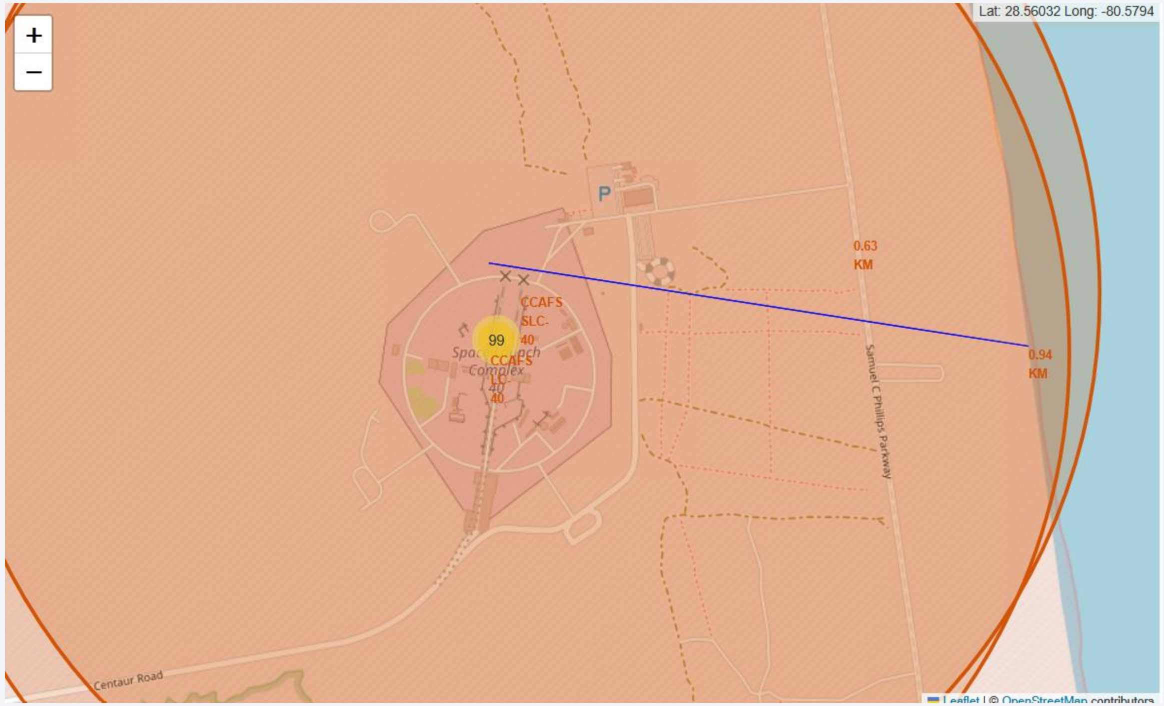
Color-labeled Launch Outcomes Folium Map

- The folium map shows the color-labeled launch outcomes on the map, with red for unsuccessful outcomes and green for successful outcome
- Color-labeled markers in marker clusters allow you to easily identify which sites have relatively high success rates.
- For example, the pictured site has a relatively low success rate with many more red markers (meaning unsuccessful outcomes) than green



PolyLines and Proximities Folium Map

- The generated folium map shows the distance calculated and displayed from a selected launch site to its proximities, in this case a coastline
- Using PolyLines on the map allows you to understand what geographic features are significant to launch sites; in this example, it is clear that proximity to the coastline is significant, as the launch site is less than 1 km from the coastline



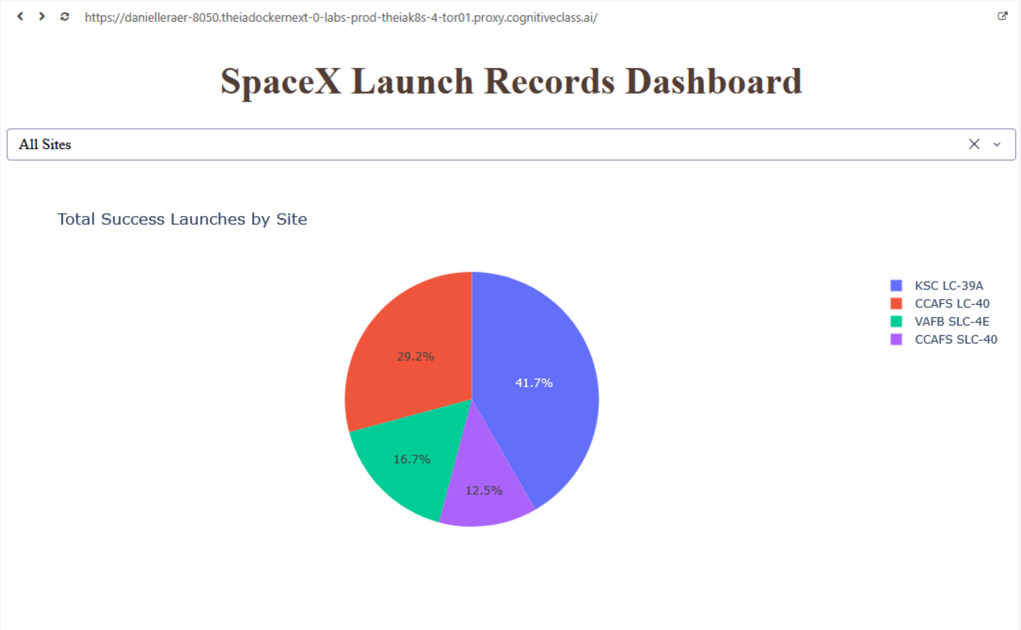
Section 4

Build a Dashboard with Plotly Dash

Launch Success Count for All Sites

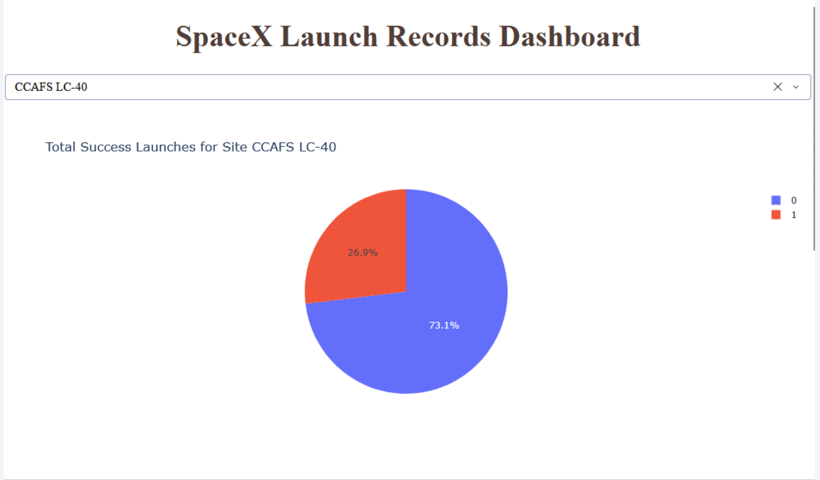
- The SpaceX Launch Records Dashboard displays a pie chart of the launch success count for all sites

- Launch site KSC LC 39A had the most successful launches with 41.7% of the total success launches

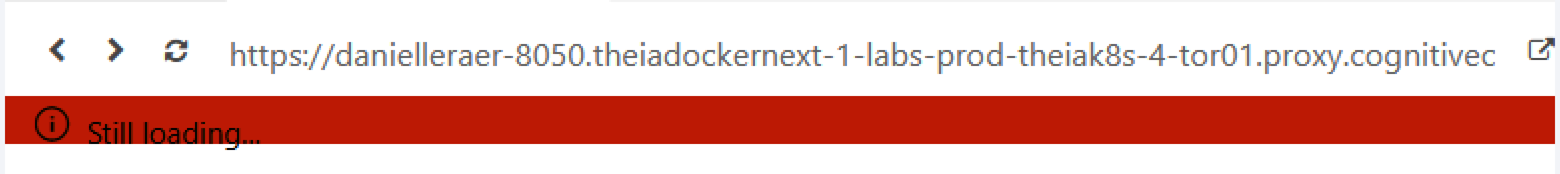


Highest Launch Success Ratio

- The dropdown can be used to filter the piechart by launch site to find the launch site with highest launch success ratio
- KSC LC 39A has the highest success ratio

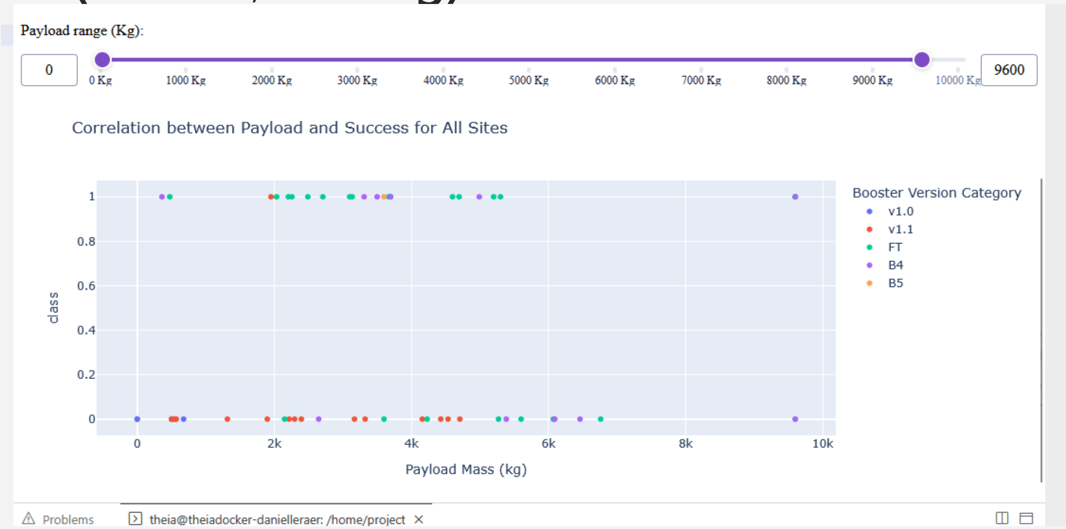


- Note that the screenshot does not show the highest launch success ratio because I did not know I would need that screenshot and I cannot recreate the dashboard now due to this persistent error:



Payload vs. Launch Outcome Scatter Plot for All Sites

- The Payload vs. Launch Outcome scatter plot shows the correlation between payload and success for all sites and can be filtered by the site dropdown and payload range slider
- The scatter plot shows that smaller payload launches (2,000 to 4,000 kg) have a higher success rate than large payload launches (over 4,000 kg)
- The booster version category with the highest success rate appears to be FT
- The booster version category with the lowest success rate appears to be V1.1



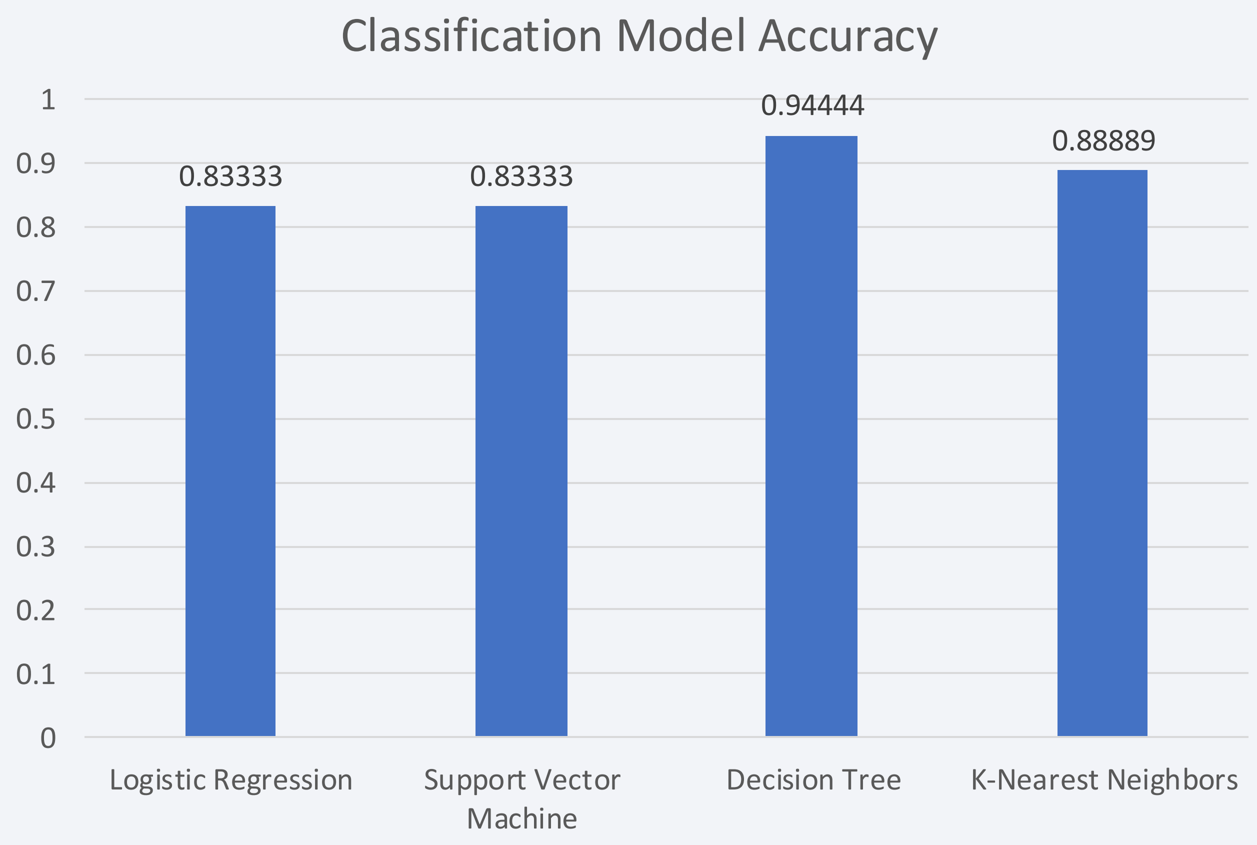
Section 5

Predictive Analysis (Classification)

Classification Accuracy

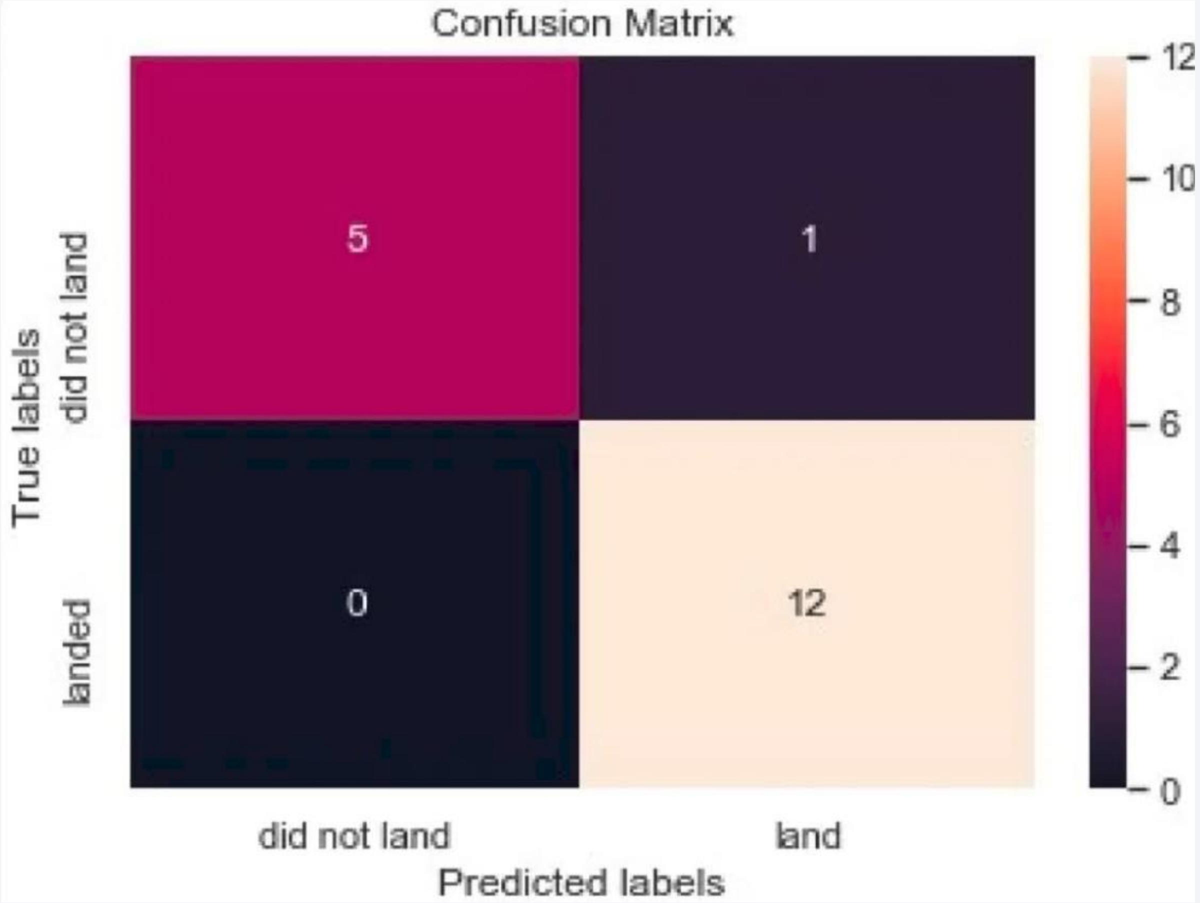
- The bar chart visualizes the built model accuracy for all built classification models

- The decision tree model has the highest accuracy



Confusion Matrix

The confusion matrix of the decision tree model shows just one false positive for land



Conclusions

- Earlier flights were less successful; as the flight number increases, the success rate increases
- While four orbit types have the highest success rate of 100%, three of those (ES-L1, GEO, and HEO) had only one flight; SSO had a success rate of 100% with five flights
- Orbit types PO, ISS, and LEO had more success with higher payloads relative to other orbit types
- Launch site KSC LC 39A had the most successful launches with 41.7% of the total success launches and the highest successful launch rate at 76.9% success
- The booster version category with the highest success rate is FT
- The Decision Tree Classifier model has the best accuracy rate at 94%

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Thank you!